1.1 Computers are electric

In this lesson

You will learn:

- what a processor is
- how computers use electricity.

Spiral back

In Student Book 1 you learned what a computer is. In this unit you will learn what is inside a computer.

What is electricity?

A **device** is something people make to do a useful task. Many devices use electricity.

Electricity is a kind of energy. Wires bring electricity. The electricity flows along the wire.

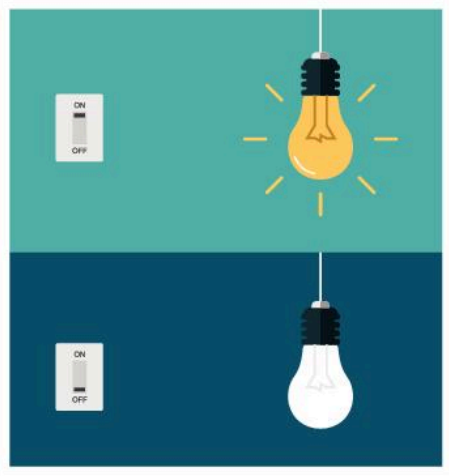
But electricity is dangerous. If you touch an electric wire you will get a bad shock. It will hurt you. So be very careful with anything that has electricity in it.



Electricity ON and OFF

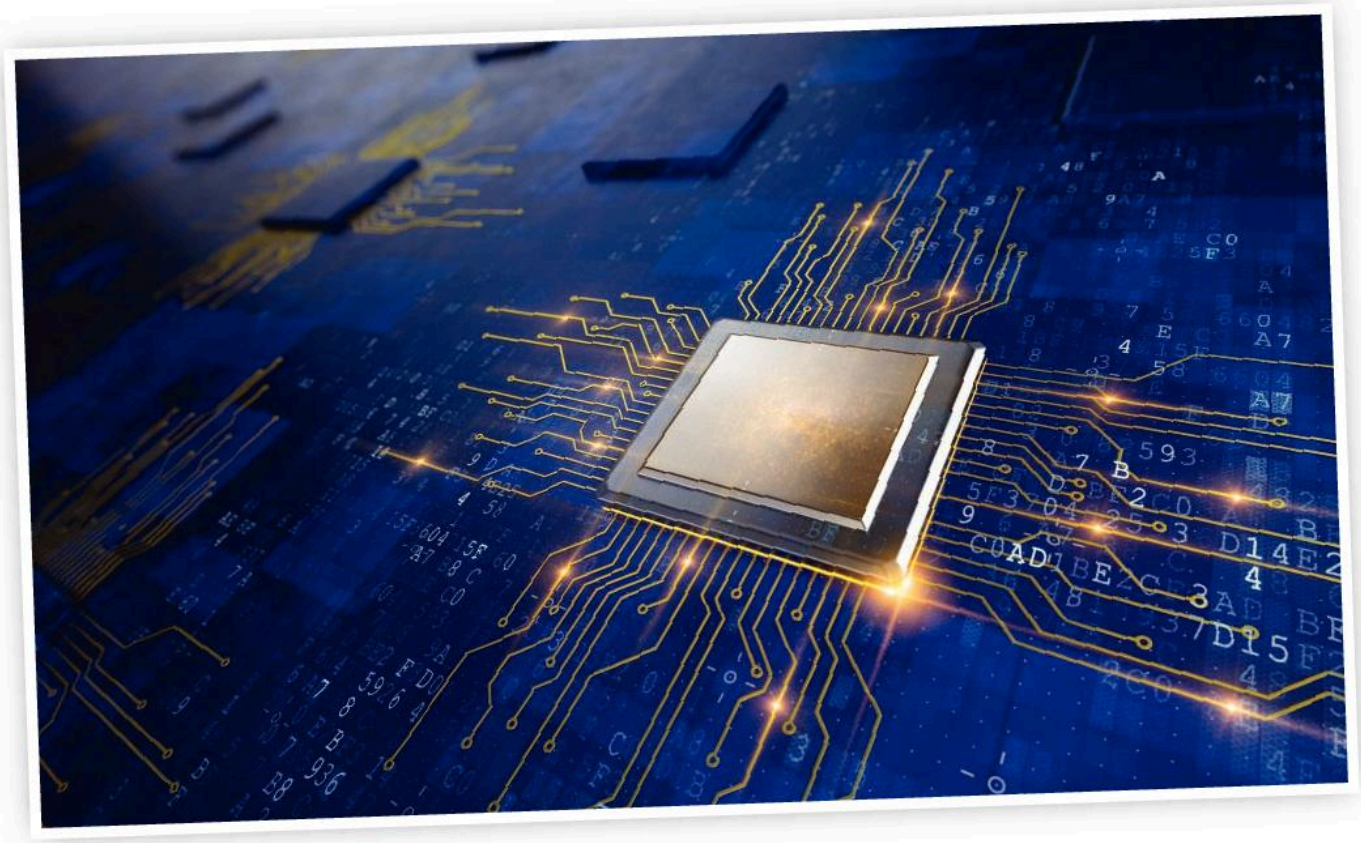
Electricity can be ON or OFF. Think of a light bulb. When the switch is ON, electricity can get through. The light goes on.

The OFF switch stops the electricity. The light goes off.



Inside the computer

There is an electrical device inside the computer. It is called a **processor**. The processor uses on/off electrical signals. Everything that happens inside the processor is made of on/off electrical signals.



Activity

Tell a partner about the last time you used a computer. What did you use it for?

Extra challenge

Draw a computer. Show the wire that brings electricity to the computer. Draw a circle where you think the processor is.

Think again

What must you do to stay safe when you use electronic devices?

1.2 Output

In this lesson

You will learn:

- what output is
- what computer parts are used for output.

Inside a computer

A computer holds lots of information. For example numbers and pictures. It holds the information using electricity.

But you cannot see the electricity. The circuits are too small. It is dangerous to open a computer when the electricity is switched on.

Output devices

You cannot look at the electricity inside a computer. How can you find out what is inside the computer?

You use an **output device**.

Output devices take the electricity from the processor. They turn the electrical signals into things you can see and use.

Output you can see

The **screen** or **monitor** of a computer is an output device. It takes information from the processor. It turns it into colours and shapes.



A monitor makes visual output. That means you can see the output.



A printer gives you output. The output is on paper. You can keep the output after the computer is switched off.

Other types of output

There are other types of output.





- **Sound output:** Speakers and headphones make sound output.
- **Movement:** A computer can make a machine move. For example, a machine can clean the floor.



Activity

What devices are used for output? Draw two different output devices. Write the names.

Extra challenge

A 3D printer makes solid objects. For example, a 3D printer can stick blobs of plastic together. See if you can find out more.

Think again

A monitor makes visual output. Say one other type of output.

1.3 Input

In this lesson

You will learn:

- what devices are used for input.

The user

A person who uses a computer is called the **user**. The user tells the computer what to do.

Everything inside the computer is made of electricity. How can the user put electricity into the computer?

Input devices

The answer is to use an **input device**. An input device takes a user's inputs. It turns them into electrical signals that the computer can understand.

Keyboard

A **keyboard** has keys. Touch a key. The keyboard sends a signal to the computer. It tells the computer which key you chose.



Mouse

A **mouse** can roll about on the desk. The mouse sends a signal to the computer. It tells the computer when it moves. A pointer on the screen moves about.

The mouse has a button. You click the button to make a choice.



Touchscreen

Some devices have a **touchscreen**. A touchscreen is used for input and output. It shows you choices on the screen. You touch the screen to make a choice.



Other input devices

There are other types of input. For example:

- A microphone turns sound into electric signals.
- A camera turns pictures into electric signals.



Activity

What devices are used for input? Draw two different devices. Write the names.

Extra challenge

Draw a device that has a touchscreen. Show yourself in the picture using the touchscreen. What is the name of the device?

Think again

What device can you use to input a song to the computer?

1.4 Joining devices together

In this lesson

You will learn:

➔ how devices are linked together.

The parts of a computer

The different devices that make a computer are called **hardware**.

The devices must be joined together.

- The input devices send signals to the processor.
- The output devices get signals from the processor.

Wired and wireless

The signals can go down wires. The signals go as electricity.

Signals can go without a wire. This is called **wireless**.

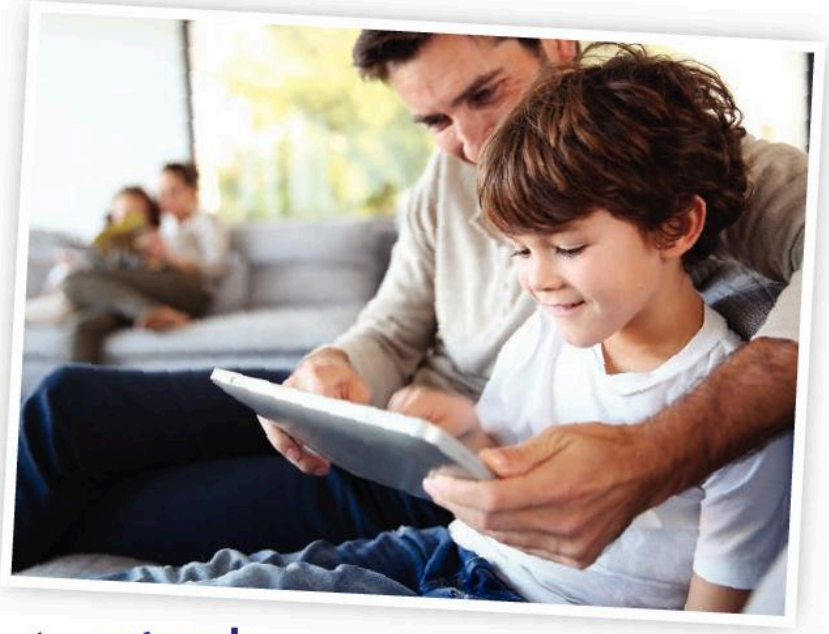




Wireless signals go through the air. They can go as radio signals. Wireless signals are safe for humans.

In the same case

Sometimes the different hardware devices are held inside the same case. The connections are hidden inside the case.



Computer networks

Computers can be joined together. The connections can be wired or wireless.

The computers can send signals to each other. This is called a computer **network**.

The **internet** is a big computer network.



Activity

These pictures show things we can do using computer devices.

Draw a picture of you using a computer. Label the devices. Show the connections.









Extra challenge

Is there a network in your school? Find out more.

Think again

Say a device that has input and output devices in the same case.

1.5 How computers can help

In this lesson

You will learn:

➔ what kinds of tasks computers can help us do.

What can computers do?

Mrs Harb is a teacher. She uses a computer to help with her work.



What tasks can people do by using a computer?

- Jobs where the same task is repeated over and over again, or needs to be done very carefully
- Work out hard mathematics very quickly
- Share information with lots of people all at once.



In the picture, computers are used to build cars.

Activity

Look at this picture.



Which jobs in the picture could a computer do? Which activities in the picture could a computer do? Write down your answers like this:

A computer can work out how much to charge for food.

Extra challenge

Can you think of any other everyday activities a computer could do?

Explore more

Is using a computer always the best way to carry out a task? Talk with an adult about tasks that humans do better than computers.

1.6 When computers can't help

In this lesson

You will learn:

- what kinds of tasks computers cannot do easily.

What can humans do better than computers?

Humans can carry out many tasks.



Humans are better than computers at:

- understanding other people
- art and creativity
- inventing new ways to do things.

Activity

These girls are doing a dance. They do not want a computer to do that task. They enjoy dancing themselves.

Think of another task that humans enjoy doing. Could computers do this task instead?

Extra challenge

In a care home, old people are looked after by kind nurses. How can a computer help the nurses? What tasks are better for the nurses to do themselves?



Digital citizen of the future

Cars have been invented that can drive by themselves. They have a computer inside. Perhaps in the future nobody will drive a car. Think about the advantages and disadvantages of this. Do you want to learn to drive when you are older?

Think again

Give one reason why we might decide not to use a computer to help with a task.